COURSE DESCRIPTION			
Program	**Textile Technology**		
Course Title	**Fibre Science**		
Course Code	**1061**	**Semester**	**I**
Type of Course	**Practicum**	**Contact Hours**	**90**
Teaching Scheme (L: T:P:R)	**3:0:3:0**	**Credits**	**4.5**
CIE Marks	**40**	**ESE Marks**	**100**

Introduction

Fibre Science is a practicum-based course designed to introduce diploma students to the classification, structure, properties, and uses of natural textile fibres. The course integrates theory with laboratory experiments to develop practical skills in fibre identification using physical, chemical, and microscopic methods. It also enables students to relate fibre properties to appropriate end-use applications through systematic observation and analysis.

Course Objectives

1	To understand the classification and fundamental properties of natural textile fibres used in the textile industry.
2	To develop practical skills in fibre identification using burning, chemical, and microscopic methods.
3	To analyse experimental observations to relate fibre properties with suitable end-use applications in textiles

Course Pre-requisite

Topic	Course code	Course Title	Semester
NIL			

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Classify textile fibres based on origin and chemical nature and identify fibres using physical and burning tests.	Understand
CO2	Explain the structure and properties of cotton fibre and apply microscopic and chemical tests to identify cotton fibre.	Apply
CO3	Describe the extraction, structure, and properties of bast fibres and differentiate jute and linen using laboratory identification methods.	Apply
CO4	Explain the features and properties of silk and wool, and apply laboratory tests to identify these animal fibres.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	-	3	-	-	-	-	-	-	-
CO2	3	-	-	3	-	-	-	-	-	-	-
CO3	3	-	-	3	-	-	-	-	-	-	-
CO4	3	-	-	3	-	-	-	-	-	-	-
Total	12	3	-	12	-	-	-	-	-	-	-
Correlation Level	3	3	-	3	-	-	-	-	-	-	-
Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"											

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
3	0	3	0	6	90	4.5	20	60	80	20	40	60	140

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus – Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	P
Module I	Fundamentals and Classification of Textile Fibres	Definition and classification of Textile Fibres. Basic Fibre Properties and Identification Methods Fibre Blends: Concept and Purpose	11	0
Module I	Fundamentals and Classification of Textile Fibres	Identification of Textile Fibres Using Burning and Chemical Solubility Tests. Identification and Approximate Percentage Determination of Component Fibres in Blended Samples	0	10
Module II	Cotton Fibre – Structure, Properties, and Identification	Classification and Varieties of Cotton Structure and Chemical Composition of Cotton Fibre Chemical Behaviour, Properties, and Uses of Cotton	11	0
Module II	Cotton Fibre – Structure, Properties, and Identification	Microscopic Identification of Cotton Fibre Chemical Reaction of Cotton Fibre with Acids and Alkalis	0	10
Module III	Bast Fibres – Jute and Linen	Features of Jute Fibres and Extraction of Jute Fibres Chemical Composition, Properties and Uses of Jute Features of Linen Fibres and Extraction of Linen Fibres. Chemical Composition, Properties and Uses of Linen.	11	0
Module III	Bast Fibres – Jute and Linen	Microscopic Identification and Chemical Reaction of Jute Fibres with Acids and Alkalis. Microscopic Identification and Chemical Reaction of Linen Fibres with Acids and Alkalis.	0	10
Module IV	Natural Protein Fibres	Silk Fibre: production, processing, properties & uses Wool Fibre: production, processing, properties & uses	12	0
Module IV	Natural Protein Fibres	Microscopic Identification and Chemical Reaction of Silk Fibres with Acids and Alkalis. Microscopic Identification and Chemical Reaction of Wool Fibres with Acids and Alkalis.	0	9

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Definition and classification of Textile Fibres.	Define textile fibres and classify them based on their origin.	L	CO1	Understand	2
Definition and classification of Textile Fibres.	Explain the classification of Textile fibres based on chemical nature.	L	CO1	Understand	2
Basic Fibre Properties and Identification Methods	List and explain the essential properties of textile fibres.	L	CO1	Understand	2
Basic Fibre Properties and Identification Methods	List and explain the desirable properties of textile fibres.	L	CO1	Understand	2
Basic Fibre Properties and Identification Methods	Discuss how different textile fibres behave when subjected to a burning test.	L	CO1	Understand	1
Basic Fibre Properties and Identification Methods	Discuss the solubility behaviour of different textile fibres in various chemicals.	L	CO1	Understand	1
Basic Fibre Properties and Identification Methods	Identify various textile fibres (Cotton, Jute, Linen, Silk, Wool, Nylon, Polyester) by observing their behaviour near flame, nature of burning, odour, and residue formed.	P	CO1	Understand	3
Basic Fibre Properties and Identification Methods	Identify different textile fibres such as cotton, jute, linen, silk, wool, nylon, and polyester by chemical solubility tests.	P	CO1	Understand	3
Fibre Blends: Concept and Purpose	Explain the terms Staple and filament fibres with examples. State the Purpose of Blending Fibres.	L	CO1	Understand	1

Fibre Blends: Concept and Purpose	Identify the component fibres present in a blended sample and determine the approximate percentage composition using selective chemical dissolution.	P	CO1	Understand	4
Module II					
Classification and Varieties of Cotton	Explain the features of cotton fibre and list the cotton-producing countries.	L	CO2	Understand	1
Classification and Varieties of Cotton	State the botanical classification of cotton.	L	CO2	Understand	1
Classification and Varieties of Cotton	Explain the types of commercial cottons and their properties	L	CO2	Understand	1
Classification and Varieties of Cotton	Explain the terms Organic cotton & BT cotton. List the Hybrid Indian Cottons and their properties.	L	CO2	Understand	1
Structure and Chemical Composition of Cotton Fibre	Explain the structure of cotton fibre with a sketch.	L	CO2	Understand	2
Structure and Chemical Composition of Cotton Fibre	Explain the Chemical composition of cotton fibre and draw the microscopical longitudinal view and cross-sectional view of Cotton fibre.	L	CO2	Understand	2
Structure and Chemical Composition of Cotton Fibre	Identify cotton fibre under a microscope and examine the peculiar features of its longitudinal structure.	P	CO2	Understand	4
Chemical Behaviour, Properties, and Uses of Cotton	State the Chemical properties of Cotton fibre. (effect of acid, alkali, water, sunlight, microorganisms)	L	CO2	Understand	1
Chemical Behaviour, Properties, and Uses of Cotton	Analyse the Chemical Reaction of Cotton Fibre with Acids.	P	CO2	Apply	3
Chemical Behaviour, Properties, and Uses of Cotton	Analyse the Chemical Reaction of Cotton Fibre Alkalis.	P	CO2	Apply	3
Chemical Behaviour, Properties, and Uses of Cotton	State the physical properties of Cotton fibre. (length, strength, fineness, colour)	L	CO2	Understand	1
Chemical Behaviour, Properties, and Uses of Cotton	List the uses of Cotton	L	CO2	Understand	1
Module III					
Features of Jute Fibres and Extraction of Jute Fibres	State the features of Jute fibre Explain the process of extracting jute fibres from the plant.	L	CO3	Understand	2
Chemical Composition, Properties and Uses of Jute	State the Chemical Composition of the jute fibre and draw the microscopical longitudinal view and cross-sectional view of the Jute fibre.	L	CO3	Understand	1
Chemical Composition, Properties and Uses of Jute	Identify Jute fibre under a microscope and examine the peculiar features of its longitudinal structure.	P	CO3	Understand	2
Chemical Composition, Properties and Uses of Jute	State the Physical (length, strength, fineness, colour)and Chemical Properties(effect of acid, alkali, water, sunlight, microorganisms) of Jute fibre. State the uses of Jute fibre.	L	CO3	Understand	2
Chemical Composition, Properties and Uses of Jute	Analyse the Chemical Reaction of Jute Fibre with Acids and Alkalis.	P	CO3	Apply	3
Features of Linen Fibres and Extraction of Linen Fibres	State the features of Linen fibre. Explain the process of extracting linen fibres from the plant.	L	CO3	Understand	3
Chemical Composition, Properties and Uses of Linen	State the Chemical Composition of the Linen fibre and draw the microscopical longitudinal view and cross-sectional view of the Linen fibre.	L	CO3	Understand	1
Chemical Composition, Properties and Uses of Linen	Identify Linen fibre under a microscope and examine the peculiar features of its longitudinal structure.	P	CO3	Understand	2

Chemical Composition, Properties and Uses of Linen	State the Physical (length, strength, fineness, colour)and Chemical Properties(effect of acid, alkali, water, sunlight, microorganisms) of Linen fibre. State the uses of Linen fibre.	L	CO3	Understand	2
Chemical Composition, Properties and Uses of Linen	Analyse the Chemical Reaction of Linen Fibre with Acids and Alkalis.	P	CO3	Apply	3
Module IV					
Silk Fibre: production, processing, properties & uses	Discuss the varieties of Silk and list the silk-producing countries	L	CO4	Understand	1
Silk Fibre: production, processing, properties & uses	Explain Sericulture & life cycle of silk worm	L	CO4	Understand	1
Silk Fibre: production, processing, properties & uses	List the Chemical Composition of raw silk. Draw the microscopic appearance of raw & degummed silk	L	CO4	Understand	1
Silk Fibre: production, processing, properties & uses	Identify Silk fibre under a microscope and examine the peculiar features of its longitudinal structure.	P	CO4	Understand	2
Silk Fibre: production, processing, properties & uses	Describe the preparation of Raw silk from cocoons.	L	CO4	Understand	2
Silk Fibre: production, processing, properties & uses	Explain the method of degumming and weighting of silk . List the Physical and Chemical properties & uses of silk	L	CO4	Understand	1
Silk Fibre: production, processing, properties & uses	Analyse the Chemical Reaction of Silk Fibre with Acids and Alkalis.	P	CO4	Apply	3
Wool Fibre: production, processing, properties & uses	List the varieties of wool and their properties	L	CO4	Understand	1
Wool Fibre: production, processing, properties & uses	Define the terms shearing and felting of wool. State the chemical composition of wool	L	CO4	Understand	1
Wool Fibre: production, processing, properties & uses	Explain the structure of wool with a sketch. Draw the microscopic appearance of Wool.		CO4	Understand	2
Wool Fibre: production, processing, properties & uses	Identify Wool fibre under a microscope and examine the peculiar features of its longitudinal structure.	P	CO4	Understand	2
Wool Fibre: production, processing, properties & uses	Explain the scouring, chlorination, and carbonization of wool.	L	CO4	Understand	1
Wool Fibre: production, processing, properties & uses	List the physical, chemical properties and uses of wool.	L	CO4	Understand	1
Wool Fibre: production, processing, properties & uses	Analyse the Chemical Reaction of Wool Fibre with Acids and Alkalis.	P	CO4	Apply	2

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Create a flowchart with written explanations tracing the journey of one natural fibre from its raw source to a finished textile product, detailing each stage of the textile value chain.	6
2	Study the characteristics of sisal fibres and prepare a brief note on their source and applications.	6
3	Write a brief note on the environmental impact of different fibre types and the need for sustainable fibres.	6
4	Study any two emerging natural fibres (e.g. banana fibre, pineapple fibre) and prepare a short note on their source and applications.	6
Module II		
5	Prepare a flow chart showing the cotton value chain from farm to finished textile product.	6
6	Find non-traditional uses of cotton (medical textiles, hygiene products) and explain why cotton is suitable.	6
7	Compare cotton garments with synthetic garments in terms of comfort, breathability, and wear experience (user-based observation).	6
8	Study documented effects of acids and alkalis on cotton fibre from reference sources and prepare a short note explaining why cotton behaves that way.	6
Module III		

9	Prepare a report on the importance of Jute in the Indian economy	6
10	Study the characteristics of Hemp fibres and prepare a brief note on their source and applications.	6
11	Collect information on major jute or linen exporting countries and their key products.	6
12	Study and make a report on why bast fibres show specific behaviour towards acids and alkalis, and relate this to their chemical composition	6
	Module IV	
13	Explore ethical silk (Ahimsa silk) or sustainable wool practices and write a short note.	6
14	Study the features of spider silk and lotus silk and prepare a report.	6
15	Prepare a report showing how silk and wool can be differentiated using microscopic features and chemical behaviour.	6
	Total Self-Learning hours	90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	TEXT BOOK OF FIBRE SCIENCE AND TECHNOLOGY	S.P.Mishra	New Age Intl.
2	TEXTILE SCIENCE	Gohl & Valensky	Mahajan publishers Ahmadabad

REFERENCE			
Sl. No.	Title	Author	Publication
1	TEXTILE SCIENCE	J T Marsh	Read Books
2	FUNDAMENTALS OF FIBRE SCIENCE	L. C. Sonntag, P. M. Joseph	John Wiley & Sons
3	HANDBOOK OF TEXTILE FIBRES	J. Gordon Cook	Woodhead Publishing

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://nptel.ac.in/courses/116102026	All
2	https://textilelearner.net/different-methods-of-textile-fibers-identification/	I

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Compound Light Microscope (Monocular / Binocular)	Compound light microscope, 10× and 40× objectives, Coarse and fine focusing, LED illumination system	1

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology – Practicum

Mode	Attendance	CA1	CE	CA2	CA3	CA4	CA5	ESE	
		Average of Self learning activities from each module	Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)	Lab Examination (internal)
Duration				3 hours	3 hours	2 hours	2 hours	2.5 hours	3 hours
Exam mark		10	50	40	40	50	50	60	40
Converted to		5	10	10	10	10	10		
Marks	5	5	10	10		10		60	40

Total	40							100
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						Total Marks
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60